

Mayank Malik

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Summary

B.Tech CSE student at UPES with strong interests in AI/ML, backend systems, and scalable web platforms. Combines technical development with leadership, operations, and stakeholder management through high-impact university initiatives. Experienced in building production-ready projects, leading teams of 100+ students, and managing execution across fast-paced technical and organizational environments.

Education

UPES Dehradun	Aug 2024 – 2028
Bachelor of Technology in Computer Science Engineering	CGPA: 8.51/10.0
Model School Sec-4, Rohtak	2024
Class XII (CBSE)	76.5%

Experience & Leadership

PR & Sponsorship Head, UPES IET Community	Feb 2026 – Present
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- Leading PR and sponsorship strategy for the UPES IET Community across online and offline platforms.
- Mentored core team members on sponsor outreach, stakeholder communication, and deal negotiation to improve execution quality and industry readiness.
- Anchored Campus X Corporate, a professional development event covering interview preparation and career strategy, securing 100+ registrations.
- Managing sponsorship acquisition pipeline including outreach, pitching, negotiations, and brand visibility operations.

PR Head, IEEE Computational Intelligence Society (CIS) Student Chapter	Apr 2026 – Present
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- Leading public relations, outreach strategy, sponsorship coordination, and visibility initiatives for the IEEE CIS Student Chapter at UPES.
- Coordinating collaborations, event promotions, and communication workflows to strengthen chapter engagement and external presence.

Lead Enabler & Senior Mentor, Career and Services Department, UPES	Sep 2025 – Present
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- Managing internship coordination and mentorship operations for approximately 100 AIML students under the SAMARTH Domain Internship Program.
- Conducted structured interviews and onboarded Enablers responsible for batch-level execution and communication.
- Mentoring first-year B.Tech students under the SRIJAN Social Internship Program while supervising execution quality, accountability, and onboarding workflows.
- Selected through direct faculty recommendation based on leadership, execution capability, and student impact.

Lead Change Maker, SRIJAN Social Internship Program	Jan 2025 – Aug 2025
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- Managed operations across 3 student batches involving 100+ students and led a team of 5 Change Makers.
- Coordinated with NGO stakeholders and university departments to ensure structured execution, communication flow, and timely internship deliverables.
- Led conflict resolution, reporting, and inter-batch coordination during the complete internship cycle.

PR & Sponsorship Executive, Manthan – The Marketing Club of UPES	Feb 2025 – Present
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- Served as Lead Point of Contact for external teams during Riwaayat-e-Prastav at UURJA Fest.
- Managed hospitality, logistics, communication, and sponsor coordination for national-level participating teams.
- Secured external registrations and contributed to visibility campaigns across campus and social platforms.

Projects

Waffle Da! Official Ordering Platform	Python, Full-Stack
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- Built and deployed a production-ready ordering platform for a live food brand featuring real-time order tracking, push notifications, and a complete admin dashboard.
- Implemented a dual-menu architecture with automatic stall activation during college events; platform has processed 100+ live orders without major operational issues.
- Delivered continuous feature iterations while maintaining system stability and UI consistency in a real-world deployment environment.

ML-Enhanced Dynamic MAC Protocol Optimization

Python, SimPy, Random Forest

- Built a continuous-time physics-aware Discrete Event Simulation framework using SimPy to model Slotted ALOHA and CSMA/CA MAC protocols under dynamic wireless network conditions.
- Engineered a Random Forest classification pipeline that dynamically selected the optimal MAC protocol in real time using collision probability, arrival rate, and node density metrics.
- Simulated RF propagation delay, spatial node distribution, bursty traffic generation, and Binary Exponential Backoff mechanisms to model real-world wireless communication behavior.
- Successfully prevented broadcast storm scenarios under high network load, reducing collision rates to approximately 0.29% during stress-test simulations involving 17,000+ packets/sec.

Delhi NCR AQI Classification Engine

Python, XGBoost, scikit-learn

- Developed a production-grade machine learning pipeline to classify Delhi NCR Air Quality Index into Good, Poor, and Bad health-risk categories using atmospheric and meteorological data.
- Improved model accuracy from approximately 57% to 98% by redesigning the data harmonization pipeline and transitioning from Random Forest to XGBoost with optimized hyperparameters.
- Built automated visualization and evaluation modules including confusion matrices, feature importance analysis, and publication-ready research graphs.
- Designed an interactive CLI inference engine capable of real-time AQI prediction, confidence scoring, and dynamic historical scenario testing.

IMC Prosperity Trading Bot

Python, Algorithmic Trading

- Improved execution quality and robustness of an algorithmic trading strategy through safer state handling, enhanced fair-value estimation, and threshold refinement mechanisms.
- Engineered a safe mid-price calculation pipeline capable of handling one-sided order books and preventing corrupted EMA state updates during volatile market conditions.
- Implemented aggressive edge filtering, refined trade thresholds, and corrected queue-priority logic to improve order execution quality and reduce low-expectation trades.
- Refactored the codebase with stronger typing, named constants, and modularized strategy flows to improve maintainability and reduce silent state corruption bugs.

Skills

Languages: Python, C, SQL, JavaScript

AI/ML: Machine Learning, Generative AI, NumPy, Pandas, Matplotlib, Prompt Engineering

Web & Tools: HTML, CSS, Git, GitHub, MS Excel

Core Concepts: Data Structures and Algorithms, Backend Systems, Stakeholder Management, Leadership, Public Speaking

Recognition

- Promoted to PR & Sponsorship Head within 1 month based on execution impact and outreach performance.
- Selected as Senior Mentor through direct faculty recommendation for leadership and operational excellence in SRIJAN.
- Elected Head Boy at Model School Rohtak, representing 1,160+ students across Nursery to Grade 12.
- Served as Class Representative for UPES B.Tech CSE batches, acting as the primary liaison between students and faculty.